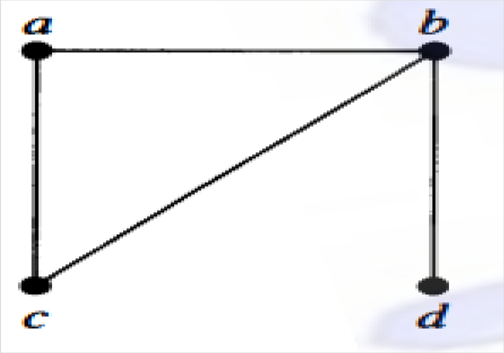
GRAPH

CSE 2213 – Discrete Mathematics

GRAPH

- A graph $G = (V, E)$ consists of V, a nonempty set of vertices (or nodes), and E, a set of edges.

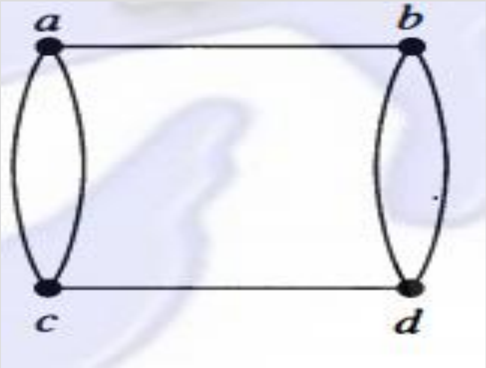


DIFFERENT TYPES OF GRAPH

- Undirected Graph
  - Simple graph
  - Multi graph
  - Pseudo graph
- Directed Graph
  - Simple directed graph
  - Directed multigraph

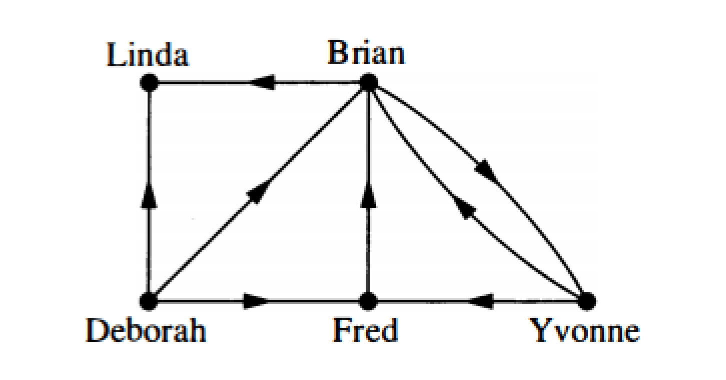
UNDIRECTED GRAPH

- Each edge connects two different vertices
  - Represented by an unordered pair of two vertices
  - An edge that connects u and v is written as $\{u, v\}$

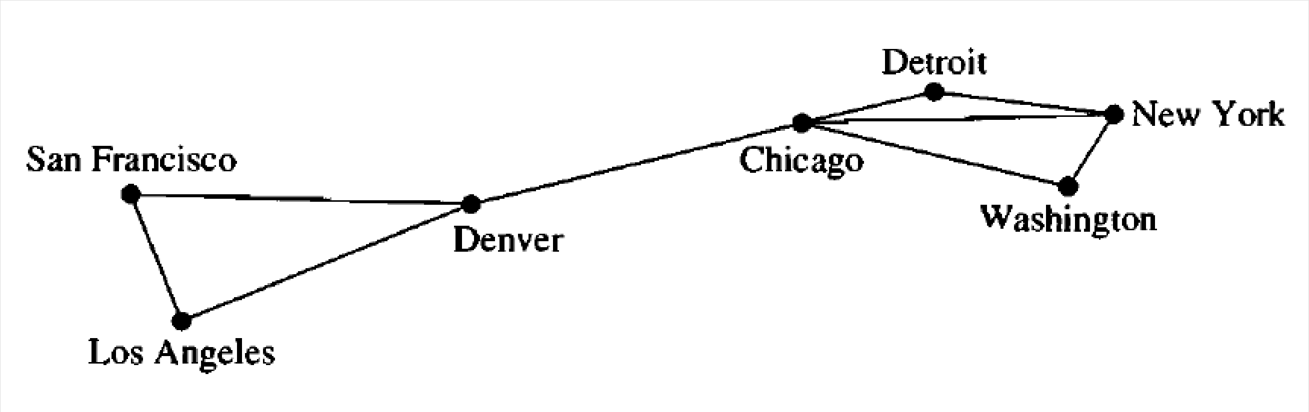


DIRECTED GRAPH

- Also known as digraph
- Each edge connects two vertices
- Each edge has a direction associated with it
  - Directed edge
  - Represented by an ordered pair of two vertices
  - If a directed edge starts at u and ends at v, then it is represented as (u, v)



A COMPUTER NETWORK

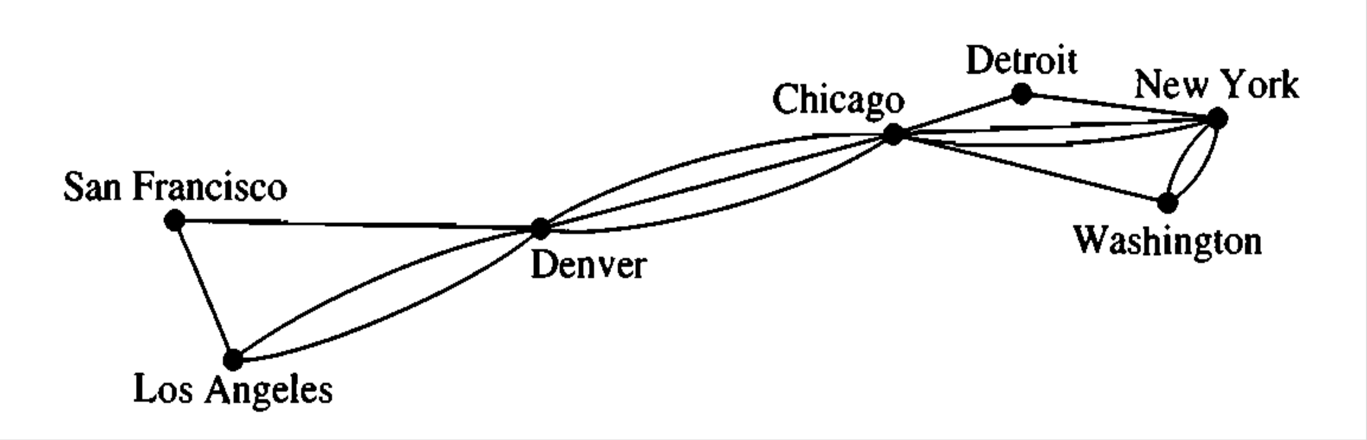


- The data centers of a company are geographically distributed
- This graph shows the connection between data centers
- What sort of graph is it?

SIMPLE GRAPH

- Each edge connects two different vertices
  - Represented by an unordered pair of two vertices
  - An edge that connects u and v is written as $\{u, v\}$
- No two edges have the same pair of vertices

A COMPUTER NETWORK

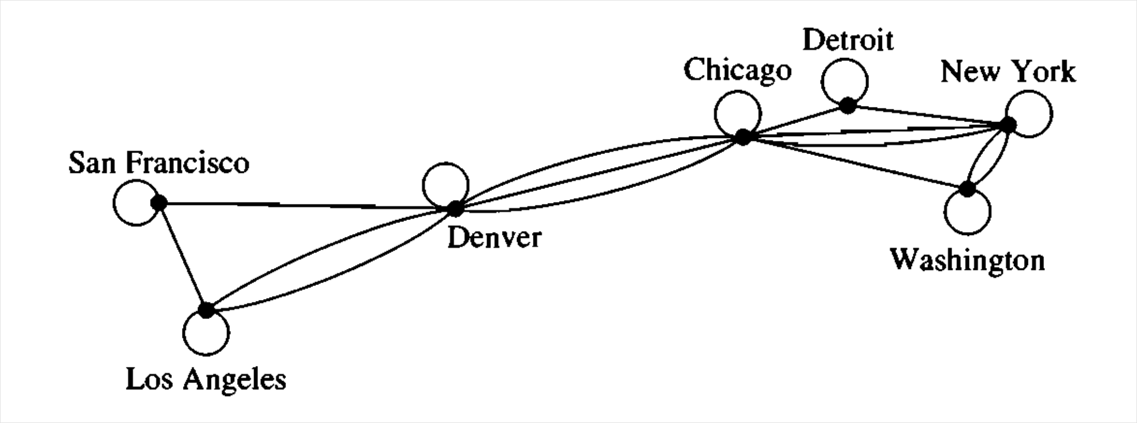


- This graph shows links between data centers
  - Note that there exists multiple links between some pair
- What sort of graph is it?

MULTIGRAPH

- Graphs that may have multiple edges connecting the same vertices are called multigraphs.

A COMPUTER NETWORK

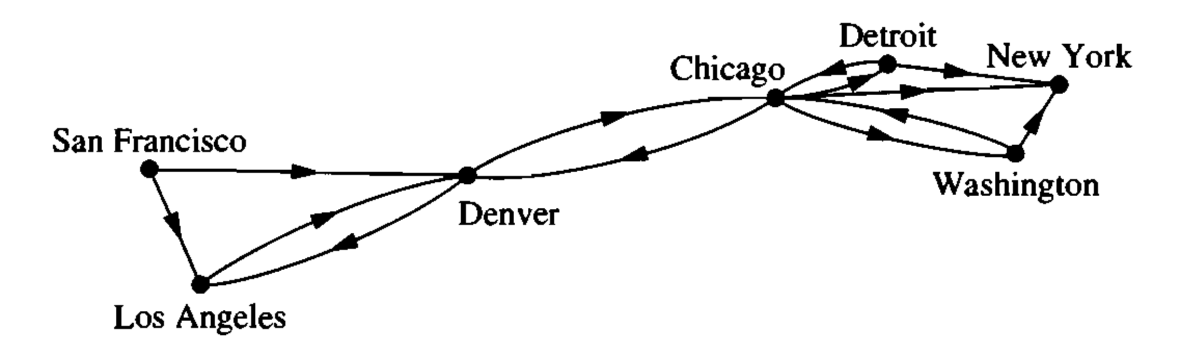


- Now there are links that connect a data center to itself!!!
- What sort of graph is it?

PSEUDOGRAPH

- Each edge connects two vertices
- There may exist edges that connect the same pair of vertices
- There may exist edges that connect a vertex to itself
  - Known as loops

A COMPUTER NETWORK



- This time the links are simple and data can be passed in one direction through a link
- What sort of graph is it?

VARIANTS OF DIRECTED GRAPH

- Simple directed graph
  - No multiple edge, no loop
- Directed multigraph
  - Multiple edge allowed, loop allowed

MIXED GRAPH

- Edges may be directed, or undirected
- Multiple edges allowed
- Loops allowed

DIFFERENT TYPES OF GRAPHS

TABLE 1 Graph Terminology.

Type	*Edges*	*Multiple Edges Allowed?*	*Loops Allowed?*
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

EXERCISE

- Chapter- 9.1 (Rosen,6th edition)
- Page-596
- Ex: 4,5,7,8,9

EXERCISE

Acquaintanceship graph

A graph that represents whether two people knows each other or not

What sort of graph is it?

EXERCISE

Hollywood graph

A graph that represents whether two actors have worked together in movies

What sort of graph is it?